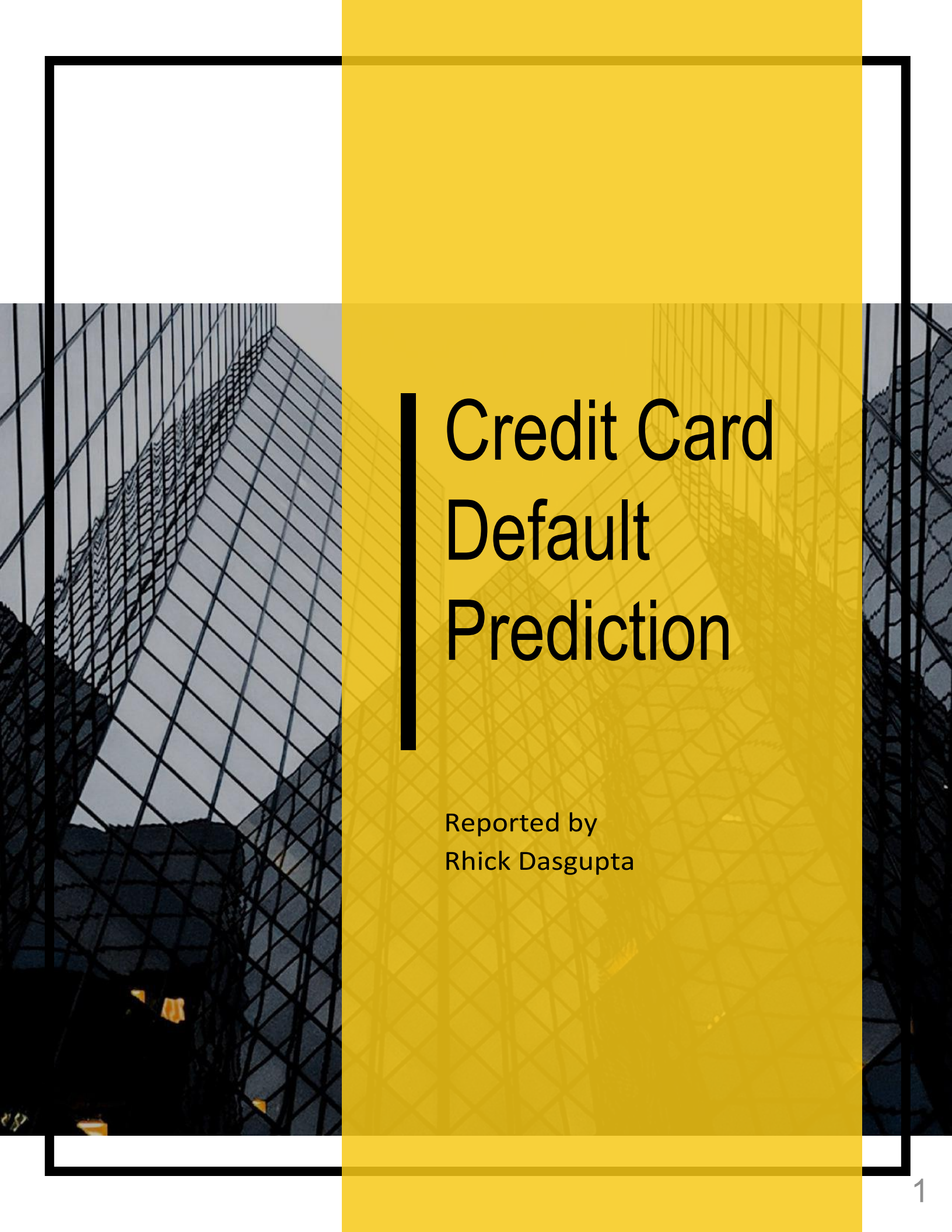




Credit Card Default Prediction

Reported by
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1. Introduction

Problem Statement

This business problem is a supervised learning example for a credit card company. The objective is to predict the probability of default (whether the customer will pay the credit card bill or not) based on the variables provided. There are multiple variables on the credit card account, purchase and delinquency information which can be used in the modelling.

PD modelling problems are meant for understanding the riskiness of the customers and how much credit is at stake in case the customer defaults. This is an extremely critical part in any organization that lends money.

Need of the study

Credit behavior assessment is a crucial aspect of the lending and financial industry. It involves evaluating an individual's or a business entity's financial habits, patterns, and history to determine their creditworthiness.

Business/Social Opportunity

- Assessing credit behavior helps financial institutions manage and mitigate the risk associated with lending money. By understanding an individual's credit history, lenders can estimate the likelihood of repayment and make informed decisions on whether to approve or deny a credit request.
- Lenders use credit reports and scores to determine the interest rates, terms, and conditions for loans. Individuals with a positive credit history are more likely to qualify for lower interest rates, while those with a poor credit history may face higher rates or may even be denied credit
- By analyzing credit behavior, lenders can predict the risk of default on credit. Individuals with a history of timely payments and responsible financial behavior are generally considered lower risk, while those with a history of missed payments or defaults may be deemed higher risk
- Credit behavior assessment provides valuable information for decision-making in various financial scenarios. It helps lenders decide the amount of credit to extend, the terms of repayment, and whether to approve or decline credit applications
- Responsible credit behavior builds trust between borrowers and lenders. Individuals or businesses that demonstrate good credit habits over time are more likely to be trusted with larger credit amounts and favorable terms
- Credit behavior assessment is not only important for lenders but also for individuals. It encourages responsible financial habits, such as timely payments, avoiding excessive debt, and maintaining a good credit score. Individuals with a positive credit history have better access to financial opportunities and can save money on interest rates
- The overall stability of the economy is influenced by the credit behavior of individuals and businesses. Sound credit practices contribute to a healthier financial system and economic stability

2. Data Report

Data Dictionary

We have been assigned with the following variables which have been collected on a real time basis over the course of various time periods within the last 2 years.

- **userid:** Unique user identification number for each customer
- **default:** Target Variable. 1 - Indicates the user has defaulted. 0 - Indicates that the person has not defaulted
- **acct_amt_added_12_24m:** The total amount of the purchases made using the credit card between 24 months in the past to the present date to the 12 months in the past to the current date.
- **acct_days_in_dc_12_24m:** The total number of days that the Credit Card Account has stayed in the Debt-Collection Status between 24 months in the past to the present date to the 12 months in the past to the current date. . Note: Debt-Collection Status: If a Customer has not even paid a minimum amount of the bill, then the account goes into a state called as debt-collection wherein the previous dues from the customer needs to be collected using an agency.
- **acct_days_in_rem_12_24m:** The total number of days that the Credit Card Account has stayed in the Reminder Status between 24 months in the past to the present date to the 12 months in the past to the current date. Note: Reminder Status: If a Customer has not yet paid the Credit Card Bill even after the last due date, the bank used to send reminders for making the payment. If an account starts receiving reminder messages, then it goes to the reminder status.
- **acct_days_in_term_12_24m:** The total number of days that the Credit Card Account has stayed in the Termination Status between 24 months in the past to the present date to the 12 months in the past to the current date. Note: Termination Status: If a Customer has paid the Credit Card Bill even after multiple reminders, then his card gets terminated and he will not be able to make any transactions using the credit card unless it gets activated again.
- **acct_incoming_debt_vs_paid_0_24m:** The ratio of the amount collected out of the total debt in an account by an agency to the total debt amount of the account in the previous 24 months from the current date.
- **acct_status:** The current status of the account. 1 represents active account, while 0 represents inactive account.
- **acct_worst_status_0_3m:** The total number of days that the Credit Card Account has stayed in the Worst Status between 3 months in the past to the present date.
- **acct_worst_status_12_24m:** The total number of days that the Credit Card Account has stayed in the Worst Status between 24 months in the past to the present date and 12 months in the past to the current date .
- **acct_worst_status_3_6m:** The total number of days that the Credit Card Account has stayed in the Worst Status between 6 months in the past to the present date and 3 months in the past to the current date .
- **acct_worst_status_6_12m:** The total number of days that the Credit Card Account has stayed in the Worst Status between 12 months in the past to the present date and 6 months in the past to the current date .
- **age:** Age of the customer
- **avg_payment_span_0_12m:** The average payment span that the customer has taken in days after the credit card bill got generated in the last one year.

2. Data Report

Data Dictionary

- avg_payment_span_0_3m: The average payment span that the customer has taken in days after the credit card bill got generated in the last three months.
- merchant_category: The category of the merchant.
- merchant_group: The group of the merchant.
- has_paid: Whether the customer has paid the current credit card bill or not. True - Paid. False - Unpaid.
- max_paid_inv_0_12m: The maximum credit card bill amount that has been paid by the customer in the last one year.
- max_paid_inv_0_24m: The maximum credit card bill amount that has been paid by the customer in the last two years.
- name_in_email: Name of the customer in email.
- num_active_div_by_paid_inv_0_12m: Ratio of the number of unpaid bills to the paid bills in the last one year.
- num_active_inv: Number of the active invoices (unpaid bills)
- num_arch_dc_0_12m: number of archived purchases that were in debt collection status in the last one year
- num_arch_dc_12_24m: number of archived purchases that were in debt collection status between 24 months in the past to the present date and 12 months in the past to the current date .
- num_arch_ok_0_12m: number of archived purchases that were paid in the last one year.
- num_arch_ok_12_24m: number of archived purchases that were paid between 24 months in the past to the present date and 12 months in the past to the current date .
- num_arch_rem_0_12m: number of archived purchases that were in the reminder status in the last one year.
- status_max_archived_0_6_months: maximum number of times the account was in archived status in the last 6 months.
- status_max_archived_0_12_months: maximum number of times the account was in archived status in the last one year.
- status_max_archived_0_24_months: maximum number of times the account was in archived status in the last two years.
- recovery_debt: The total amount that has been recovered out of the entire debt amount on the account.
- sum_capital_paid_acct_0_12m: sum of principal balance paid on account in the last one year.
- sum_capital_paid_acct_12_24m: sum of principal balance paid on account between 24 months in the past to the present date and 12 months in the past to the current date .
- sum_paid_inv_0_12m: The total amount of the paid invoices in the last one year.
- time_hours: The total hours spent by the customer in purchases made using the credit card.

2. Data Report

Visual Inspection

- The given data has 36 variables/columns and 99976 records/rows.
- We have computed Mean, Std Deviation and quantiles of all numerical variables.
- We could observe null values to be present in most of the variables which needs to be dealt with.

Variable Attributes

- Out of the 36 variables in the given dataset, 30 are float type, 3 integer type and 3 object type

	count	mean	std	min	25%	50%	75%	max
userid	99976.0	2.998977e+08	1.154177e+08	1.000053e+08	2.000267e+08	2.998832e+08	4.000641e+08	4.999888e+08
default	88976.0	1.431493e-02	1.187862e-01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00
acct_amt_added_12_24m	99976.0	1.225515e+04	3.548148e+04	0.000000e+00	0.000000e+00	0.000000e+00	4.937250e+03	1.128775e+06
acct_days_in_dc_12_24m	88140.0	2.230429e-01	5.808117e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	3.650000e+02
acct_days_in_rem_12_24m	88140.0	5.044622e+00	2.286397e+01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	3.650000e+02
acct_days_in_term_12_24m	88140.0	2.868958e-01	2.929910e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	9.700000e+01
acct_incoming_debt_vs_paid_0_24m	40861.0	1.331292e+00	2.648230e+01	0.000000e+00	0.000000e+00	1.520819e-01	6.629522e-01	3.914000e+03
acct_status	45803.0	1.042168e+00	2.027134e-01	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	4.000000e+00
acct_worst_status_0_3m	45803.0	1.172905e+00	4.201422e-01	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	4.000000e+00
acct_worst_status_12_24m	33215.0	1.337348e+00	5.750430e-01	1.000000e+00	1.000000e+00	1.000000e+00	2.000000e+00	4.000000e+00
acct_worst_status_3_6m	42274.0	1.185291e+00	4.433091e-01	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	4.000000e+00
acct_worst_status_6_12m	39626.0	1.253142e+00	5.055586e-01	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	4.000000e+00
age	99976.0	3.601628e+01	1.300131e+01	1.800000e+01	2.500000e+01	3.400000e+01	4.500000e+01	1.000000e+02
avg_payment_span_0_12m	76140.0	1.797147e+01	1.275107e+01	0.000000e+00	1.080000e+01	1.490909e+01	2.100000e+01	2.600000e+02
avg_payment_span_0_3m	50671.0	1.498979e+01	1.029742e+01	0.000000e+00	8.400000e+00	1.300000e+01	1.828571e+01	8.700000e+01
has_paid	88942.0	8.657552e-01	3.409170e-01	0.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00	1.000000e+00
max_paid_inv_0_12m	88942.0	9.362817e+03	1.367246e+04	0.000000e+00	2.390000e+03	6.170000e+03	1.140000e+04	2.790000e+05
max_paid_inv_0_24m	88942.0	1.141973e+04	1.543175e+04	0.000000e+00	3.685000e+03	7.720000e+03	1.386500e+04	5.385000e+05
num_active_div_by_paid_inv_0_12m	70051.0	1.121471e-01	2.881202e-01	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e-01	9.000000e+00
num_active_inv	88942.0	6.259922e-01	1.610536e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.000000e+00	4.700000e+01
num_arch_dc_0_12m	88942.0	6.261384e-02	3.821130e-01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.700000e+01
num_arch_dc_12_24m	88942.0	5.922961e-02	3.623228e-01	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.300000e+01
num_arch_ok_0_12m	88942.0	7.786535e+00	1.674287e+01	0.000000e+00	0.000000e+00	3.000000e+00	8.000000e+00	2.610000e+02
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num_arch_rem_0_12m	88942.0	4.840008e-01	1.395818e+00	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	4.200000e+01
status_max_archived_0_6_months	88942.0	8.217490e-01	7.166425e-01	0.000000e+00	0.000000e+00	1.000000e+00	1.000000e+00	3.000000e+00
status_max_archived_0_12_months	88942.0	1.074194e+00	7.763857e-01	0.000000e+00	1.000000e+00	1.000000e+00	2.000000e+00	5.000000e+00
status_max_archived_0_24_months	88942.0	1.248207e+00	8.205117e-01	0.000000e+00	1.000000e+00	1.000000e+00	2.000000e+00	5.000000e+00
recovery_debt	88942.0	3.601965e+00	1.162115e+02	0.000000e+00	0.000000e+00	0.000000e+00	0.000000e+00	1.641100e+04
sum_capital_paid_acct_0_12m	88942.0	1.086038e+04	2.663074e+04	0.000000e+00	0.000000e+00	0.000000e+00	8.980750e+03	5.714750e+05
sum_capital_paid_acct_12_24m	88942.0	6.615020e+03	1.924390e+04	0.000000e+00	0.000000e+00	0.000000e+00	1.027500e+02	3.418590e+05
sum_paid_inv_0_12m	88942.0	4.103637e+04	9.459685e+04	0.000000e+00	3.396500e+03	1.705750e+04	4.573950e+04	2.962870e+06
time_hours	88942.0	1.534182e+01	5.030643e+00	2.777778e-04	1.163194e+01	1.580833e+01	1.955417e+01	2.399972e+01

3. Exploratory Data Analysis

Variable Transformation

- We transform default and has_paid variables from integer type to object type.
- After the outlier treatment, 'acct_days_in_dc_12_24m', 'acct_days_in_rem_12_24m', 'acct_days_in_term_12_24m', 'num_arch_dc_0_12m', 'num_arch_dc_12_24m', 'num_arch_rem_0_12m', 'status_max_archived_0_12_months', 'recovery_debt', 'sum_capital_paid_acct_12_24m' are the variables bearing only one value. Hence these variables are also dropped and finally we are left with 14 numerical variables.

Univariate Analysis

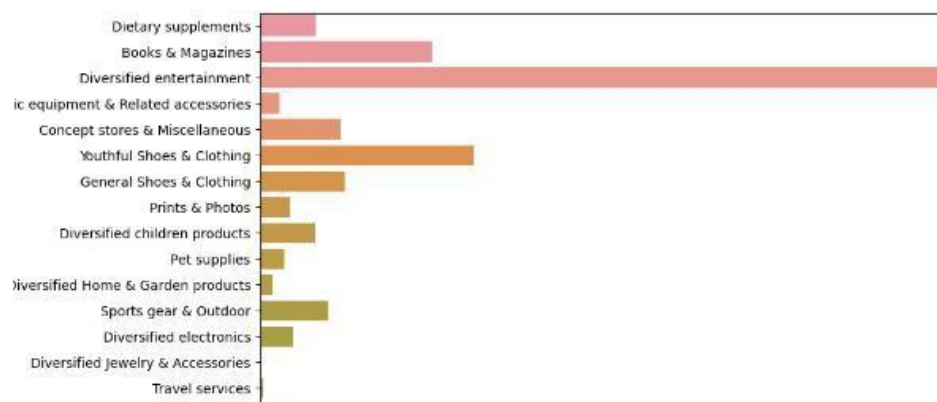
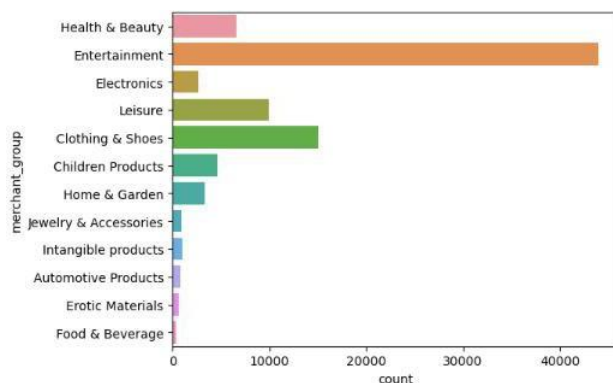
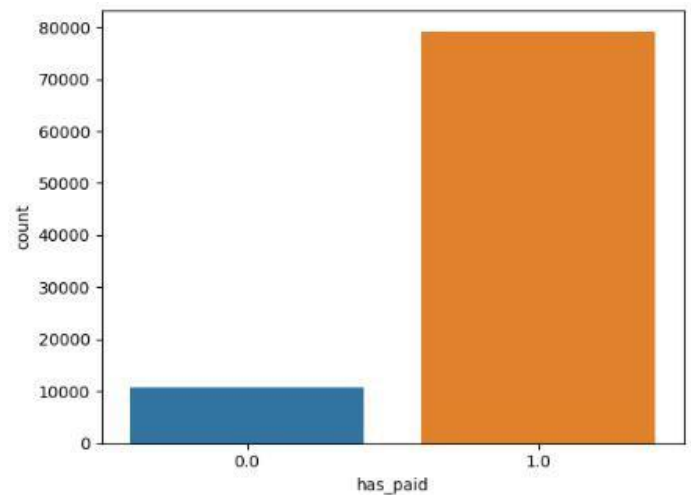
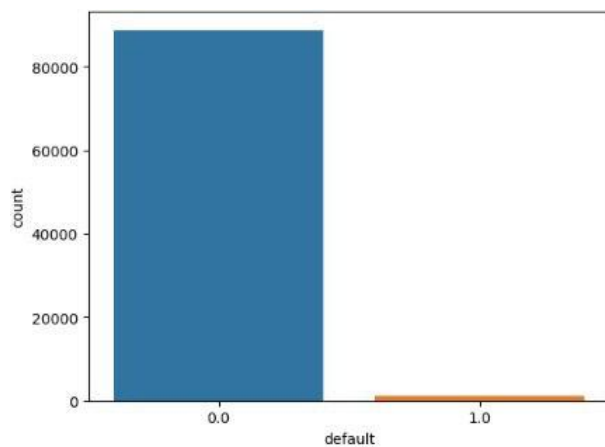


Figure 1: Univariate Analysis for categorical variables

3. Exploratory Data Analysis

- As we can see, the data is highly imbalanced with 99% of the customers being categorized as non defaulters
- 11% of the customers have paid the last bill
- Entertainment is the most frequently used merchant group.
- Diversified Entertainment is the most frequently used merchant category.

Age vs Default

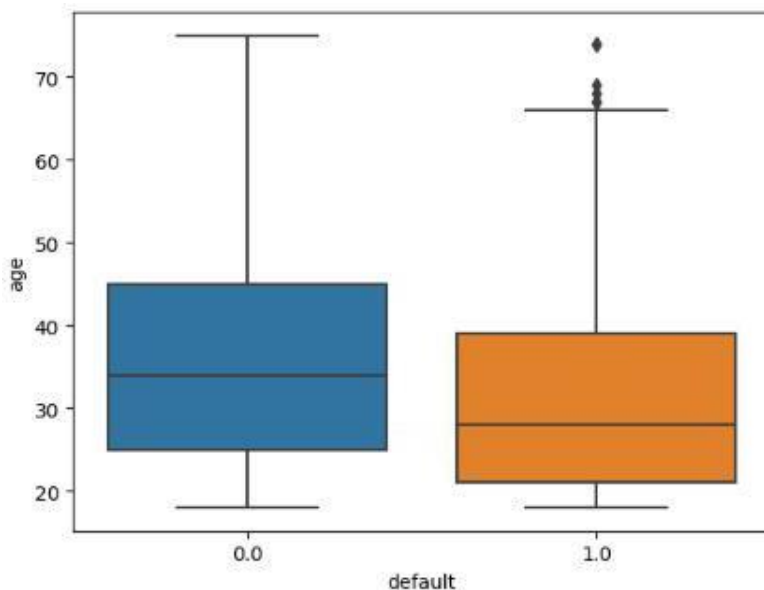


Figure 2: Age vs default

- As we can see, average of defaulters are less compared to non defaulters.
- Average age for defaulters is below 30 whereas for non defaulters average age is around 35.

Average payment span vs Default

- As we can see, payment span for defaulters is around 18 days whereas for non defaulters it is around 15 days.
- Hence, defaulters tend to take more time to settle their bills.

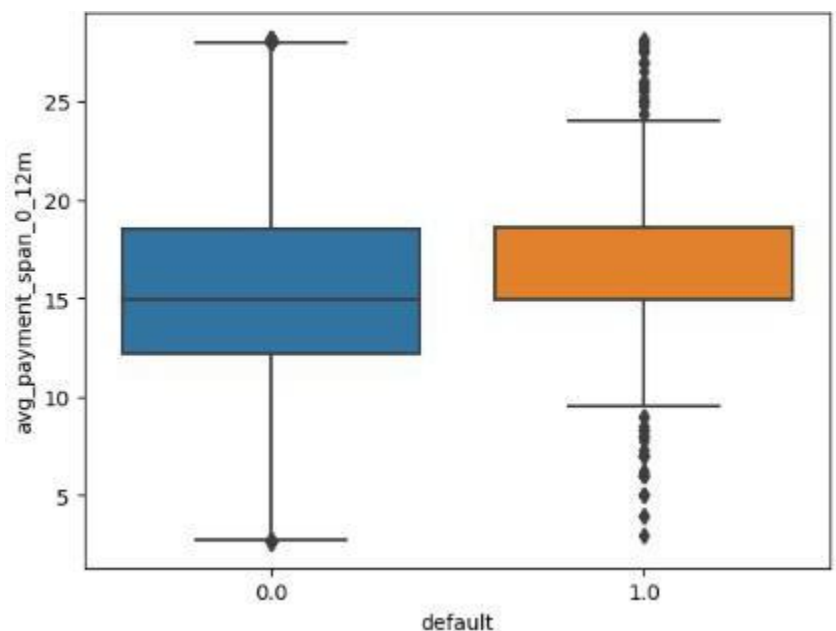


Figure 3: payment span vs default

3. Exploratory Data Analysis

Last bill status vs Default

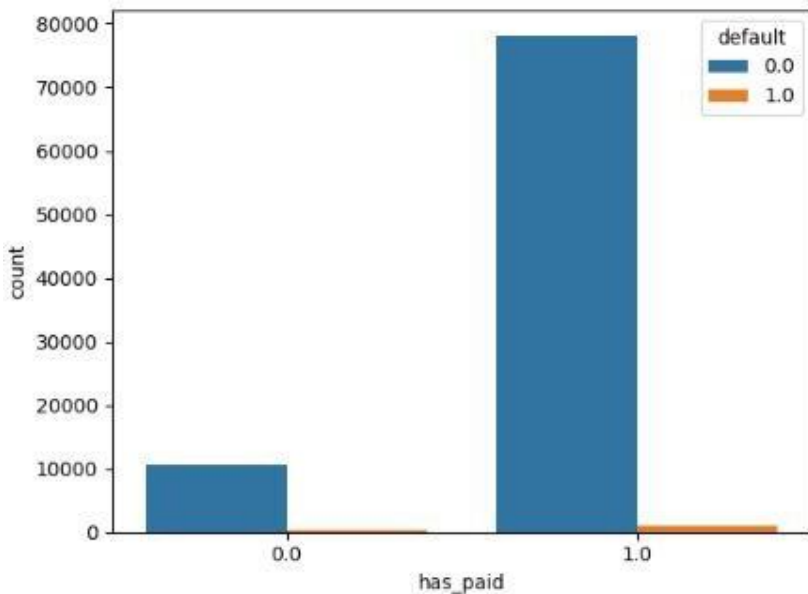


Figure 4: has_paid vs default

- For defaulters, 20% of the customers have not paid the last bill
- For non-defaulters, 11% have not paid the last bill

Maximum paid invoice vs Default

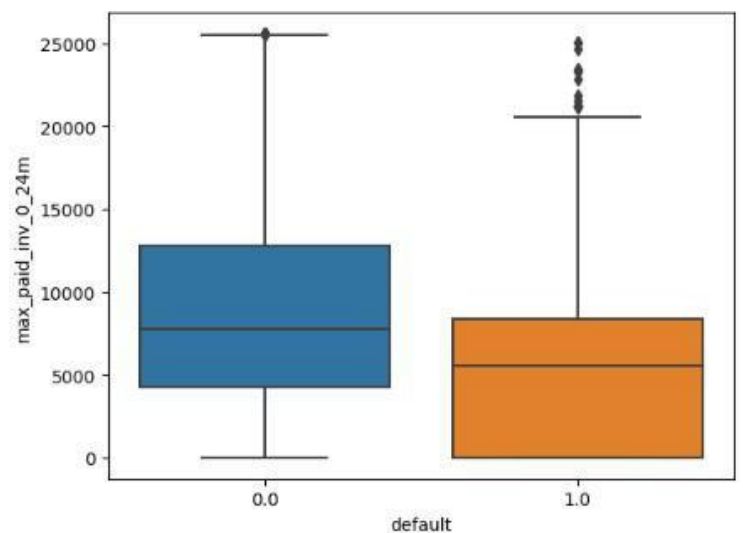
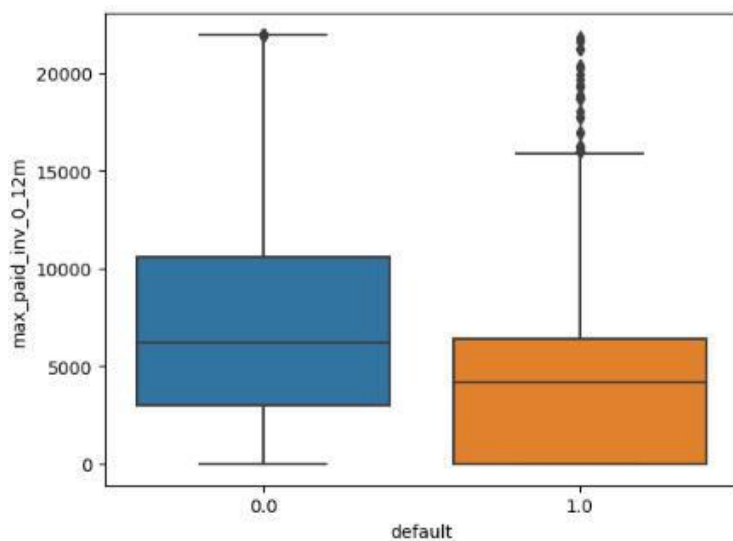


Figure 5: Maxpaid invoice in last 12 and 24 months vs default

- Maximum paid invoice for both last 12 months and last 24 months has been higher for non-defaulters compared to defaulters.

3. Exploratory Data Analysis

Merchant Group vs Default

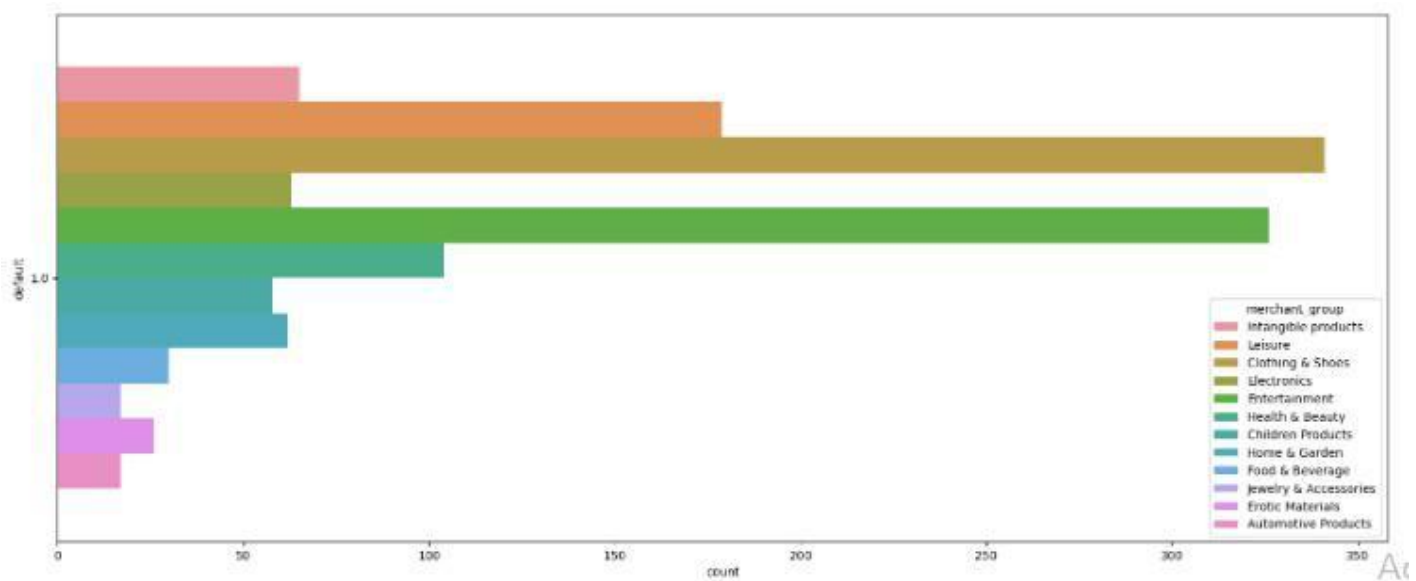


Figure 6: Merchant Group vs default

- Clothing & Shoes and Entertainment are the most defaulted merchant groups to be transacted.

Merchant Category vs Default

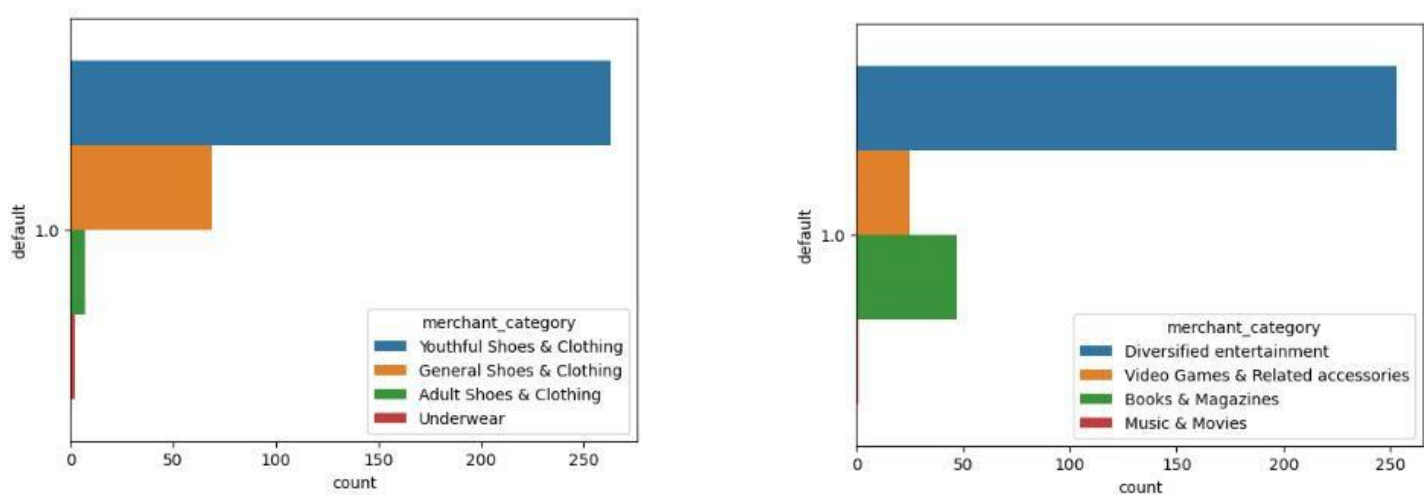


Figure 7: Merchant Category vs default

- For Clothing & Shoes group, Youthful Shoes & Clothing is the most defaulted merchant category
- For Entertainment group, Diversified entertainment is the most defaulted merchant category

3. Exploratory Data Analysis

Number of archived purchases paid vs Default

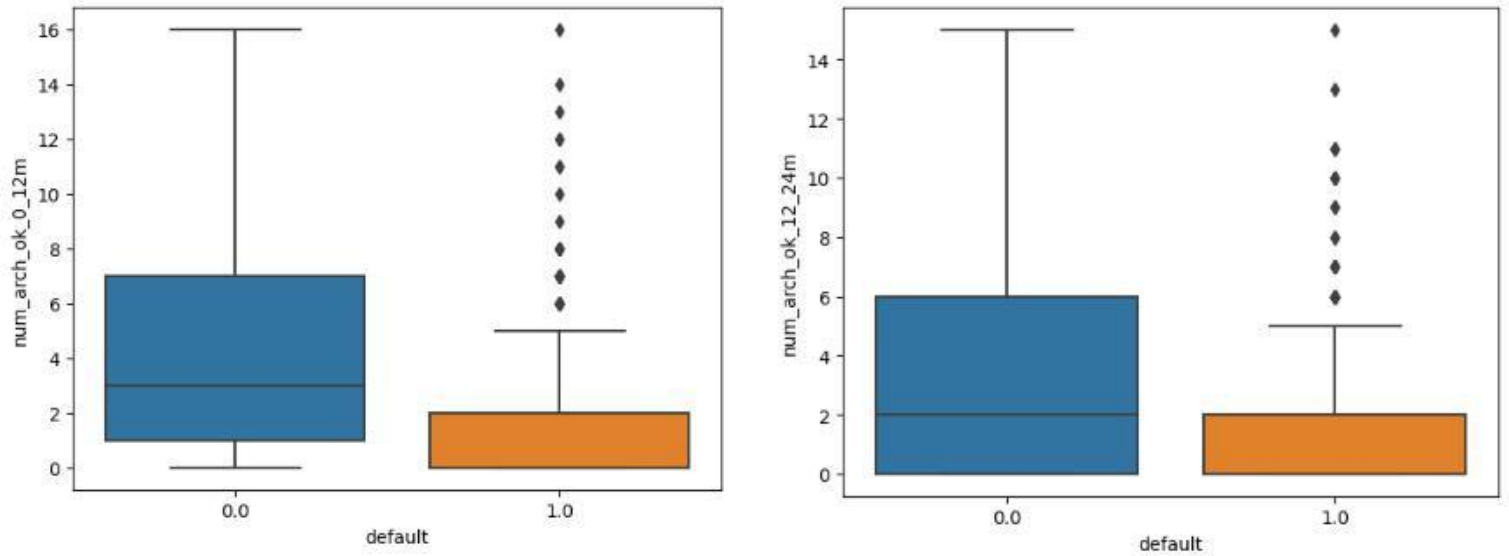


Figure 8: Number of archived purchases paid vs default

- For both last year and last to last year, non-defaulters have been more likely to pay for archived purchases than defaulters.

Paid invoices vs Default

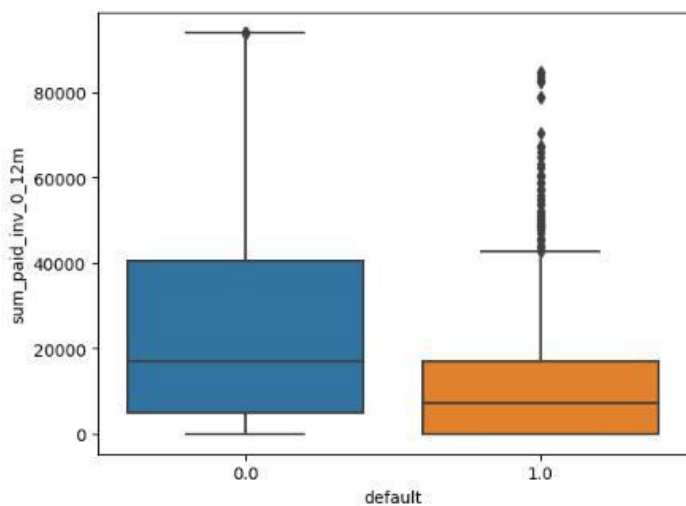


Figure 9: Invoice paid vs default

- Amount of invoices paid over the last 1 year has been higher for non-defaulters than defaulters.
- Amount of invoices paid by non-defaulters is nearly 20000 over the last 1 year whereas, for defaulters it is around 10000.

3. Exploratory Data Analysis

Data Imbalance

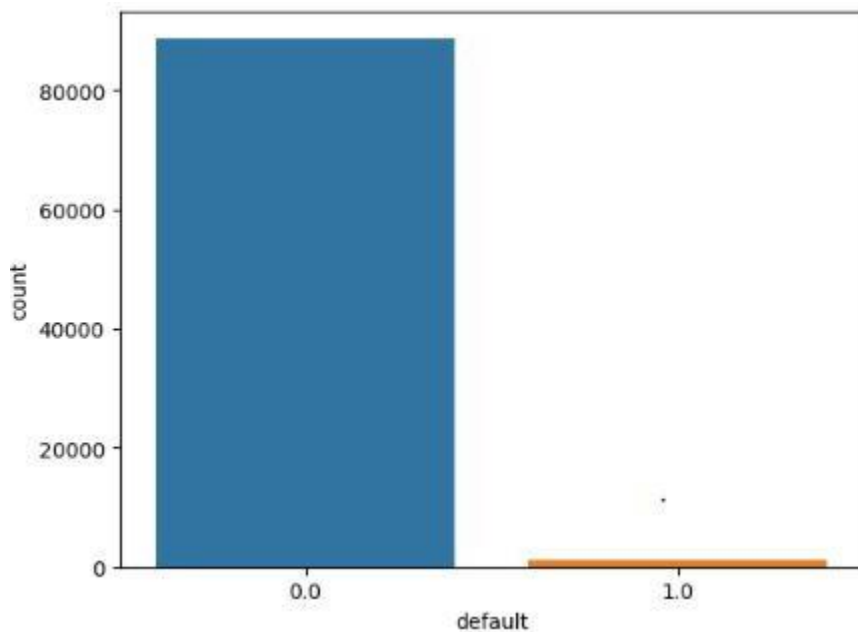


Figure 10: Data Imbalance

- The data is highly imbalanced in favour of non-defaulters.
- As we can see, 99% of the records are of non-defaulters.
- Any predictive model built upon such an imbalanced dataset will be highly biased in favour of non-defaulters.
- In other words, the model will tend to categorize anything and everything fed to it as non-defaulters.
- There can be a significant business cost associated with it if a defaulter is wrongly classified as a non-defaulter.
- Hence, before building any model, we need to pass the model through some balancing algorithm like SMOTE.

Business Insights

- The dataset has a lot of missing data. Around 8 variables had to be dropped because they have more than 30% missing values. The company should focus on revisiting the dataset to address these gaps as dropping these variables might come with losing out valuable information.
- Entertainment and Clothing & Shoes are the top 2 transacted merchant groups.
- However, the same 2 groups are the top two defaulted groups also.
- Defaulters are more likely to default on recent activities as well. As we can see, 20% of defaulters have defaulted on last bill whereas for non-defaulters only 11% have defaulted on last bill.
- Younger people are more likely to default compared to older people as we can see average age for defaulters is below 30 whereas for non-defaulters average age is around 35.
- Moreover, in the recent past (last 12 months), defaulters have taken more time to settle their bills.
- Maximum paid invoice across last 2 years has always been more for non-defaulters indicating better financial health for non-defaulters as they can afford to settle higher bill amounts.
- Non-defaulters are more likely to settle for archived purchases compared to defaulters.
- Total bill paid by non-defaulters has been higher than defaulters in the last 12 months.

3. Exploratory Data Analysis

Clustering

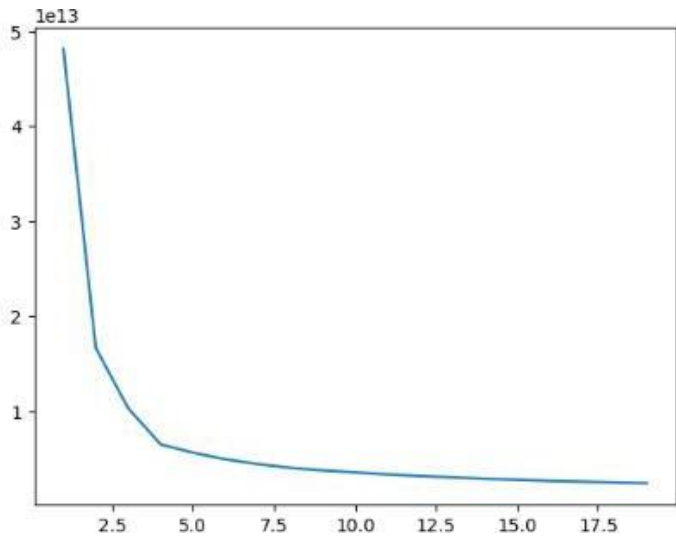


Figure 11: Elbow plot

- We use K-Means Clustering technique to conduct an unsupervised clustering for the independent variables.
- As we can see, within sum of squares or variability within clusters have reduced significantly for cluster count of upto 4 and beyond that WSS has not reduced by significant margin.
- Hence, we go ahead with 4 clusters for our clustering exercise.

Business Insights from Clustering

From our clustering exercise, we could come up with 4 clusters. The clusters and their significance are explained below:

- Cluster 1 & 2: Highly affordable segment who have made higher amount of purchases using credit card and did not miss the last bill.
- Cluster 0: This segment has not transacted as high as cluster 1 & 2 using credit cards but they have settled the last bill.
- Cluster 3: This is the most risky cluster as we can see 37% has not paid the last bill and also their transaction using credit card has been low.

4. Data Cleaning and Pre-processing

Removal of unwanted variables

- userid and name_in_email do not carry any meaningful insight pertaining to our study. Hence we drop these 2 variables.
- There are a lot of null/missing values in the dataset. Variables with more than 30% missing values are dropped as imputing them will lead to unreliable results.

Missing Value Treatment

- 'default' is the target variable for our study having 11% missing values. We drop the missing values for default column. Imputing them will make our results biased towards that particular technique and we might end up selecting the inaccurate algorithm at a later stage. Dropping the missing values will let us maintain the dataset as real time as possible.
- For the independent variables, we impute for those variables where missing value percentage is less than 30%
- For quantitative variables, we impute the missing values using the median value as there are outliers in the data.
- For categorical fields, we impute the missing values using the mode value.

Outlier Treatment

- There are outliers present in almost all of the quantitative variables.
- We impute the outliers using Inter Quartile Range (IQR) method.
- The upside outliers are imputed by upper bound = $Q3 + 1.5IQR$ where $IQR = Q3 - Q1$
- The lower side outliers are imputed by lower bound = $Q1 - 1.5IQR$
- Here Q1, Q2 and Q3 are 1st quartile (25th percentile), median (50th percentile) and 3rd quartile (75th percentile respectively)

4. Data Cleaning and Pre-processing

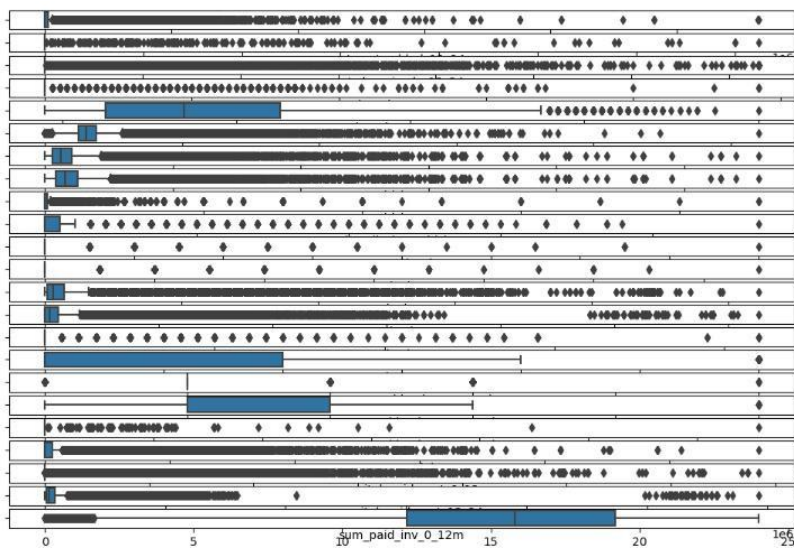


Figure 12: Box Plot before outlier treatment

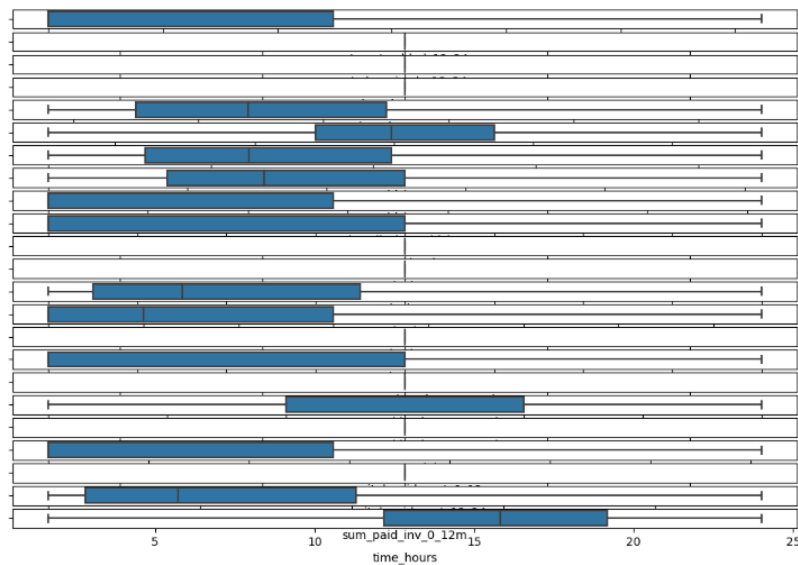


Figure 13: Box Plot after outlier treatment

Variable Transformation

Merchant Category and Merchant Group are categorical cardinal variables bearing string values which our classifier models can not process. Hence, we have done one-hot encoding for these 2 variables.

5. Model Building and Tuning

Data Pre-processing

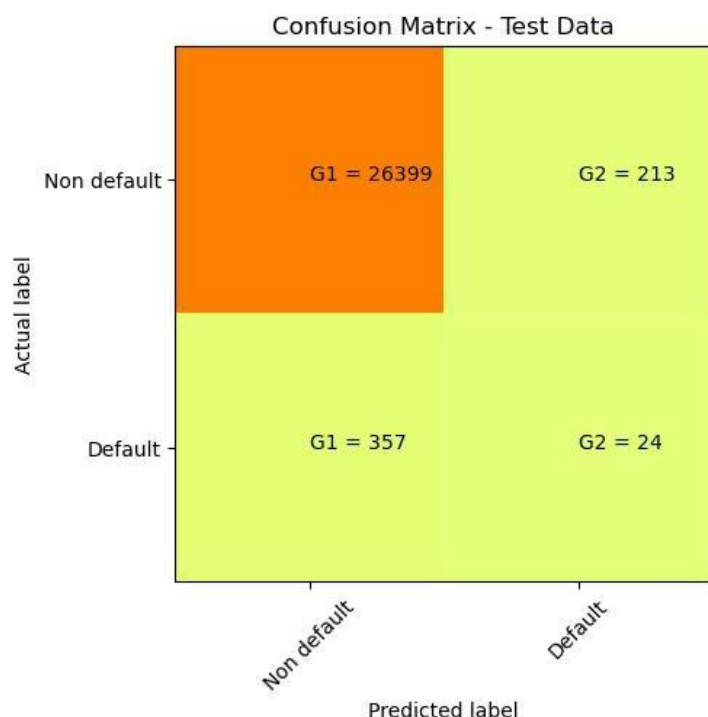
- Merchant Category and Merchant Group are categorical cardinal variables bearing string values which our classifier models can not process. Hence, we have done one-hot encoding for these 2 variables.
- Our target variable “default” is a binary variable with 0 denoting non-defaulting credit card users and 1 denoting defaulting credit card users.
- The business problem in essence is a classification problem. We use classification algorithms such as: Logistic Regression, Linear Discriminant Analysis, Gaussian Naive Bayes, Decision Tree and K-Nearest Neighbor (KNN)
- Most of the algorithms are distanced based algorithms. Hence, we have done min-max scaling on all independent variables before feeding them in the models. This is done in order to make all the independent variables as per uniform scale so that variables with larger values do not dominate variables with smaller values.

Train-Test Split

We have done a 70:30 split for training and testing dataset.

- Training set: 62983 records
- Test set: 26993 records
- There are 82 predictor variables in the final dataset

LDA Model



We have built a LDA model on the training set and evaluated performance for the same on testing set.

- Overall accuracy: 97.89%
- However, recall value is only about 6%
- Training accuracy: 97.81%
- Hence, we can infer that overfitting issue is not there but the model has not performed well on the defaulter class which is the area of interest for our study

Figure 14: LDA Confusion Matrix

5. Model Building and Tuning

Logistic Regression Model

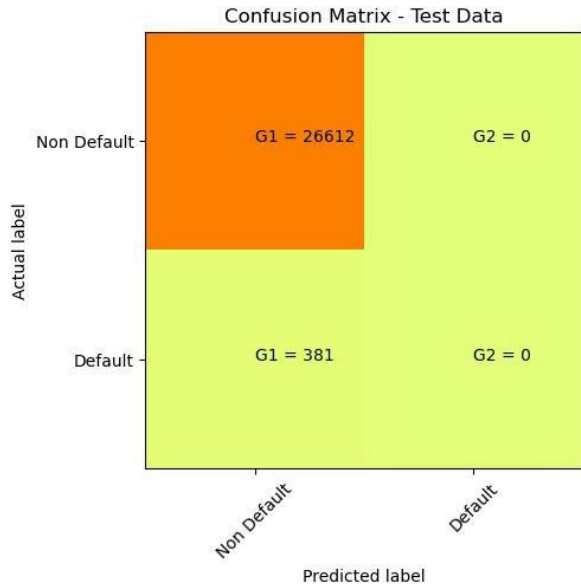


Figure 15: Logistic Regression Confusion Matrix

We have built a Logistic Regression model on the training set and evaluated performance for the same on testing set.

- Test accuracy: 98.59%
- However, recall value is 0%
- Training accuracy: 98.55%
- Hence, we can infer that overfitting issue is not there but the model has not performed well on the defaulter class which is the area of interest for our study

Gaussian Naive Bayes Model

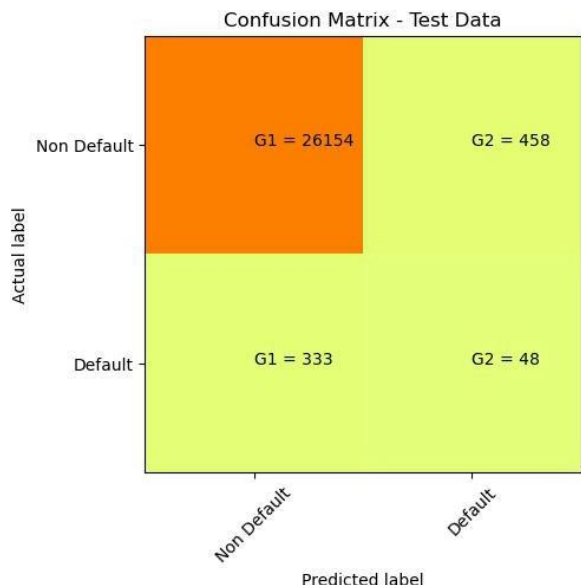


Figure 16: Gaussian NB Confusion Matrix

We have built a Gaussian Naive Bayes model on the training set and evaluated performance for the same on testing set.

- Test accuracy: 97.07%
- However, recall value is 13%
- Training accuracy: 97.09%
- Hence, we can infer that overfitting issue is not there but the model has not performed well on the defaulter class which is the area of interest for our study. However, there has been an improvement over Logistic Regression and LDA models for Defaulter class performance.

5. Model Building and Tuning

K Nearest Neighbor (KNN) Model

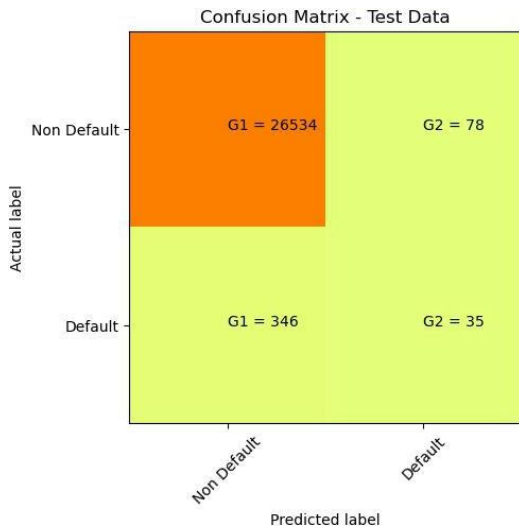


Figure 17: KNN Confusion Matrix

We have built a KNN model on the training set and evaluated performance for the same on testing set.

- Test accuracy: 98.43%
- However, recall value is 9%
- Training accuracy: 99.87%
- Hence, we can infer that overfitting issue is not there but the model has not performed well on the defaulter class which is the area of interest for our study

Decision Tree Model

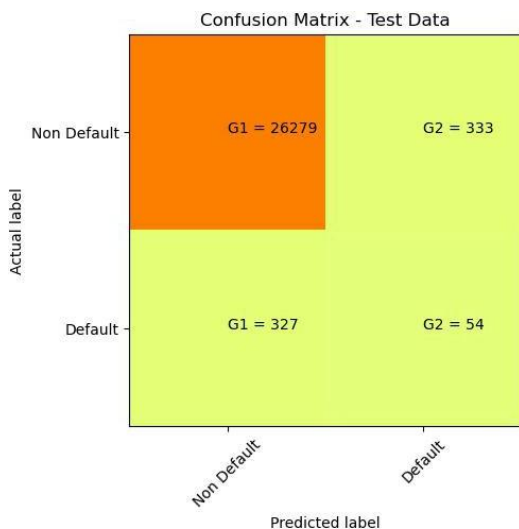


Figure 18: Decision Tree Confusion Matrix

We have built a Decision Tree model on the training set and evaluated performance for the same on testing set.

- Test accuracy: 97.55%
- However, recall value is 14%
- Training accuracy: 99.87%
- Hence, we can infer that overfitting issue is not there but the model has not performed well on the defaulter class which is the area of interest for our study. However, performance for Decision Tree model is the best among all the models under consideration for defaulter class.

5. Model Building and Tuning

Bootstrap Aggregation (Bagging)

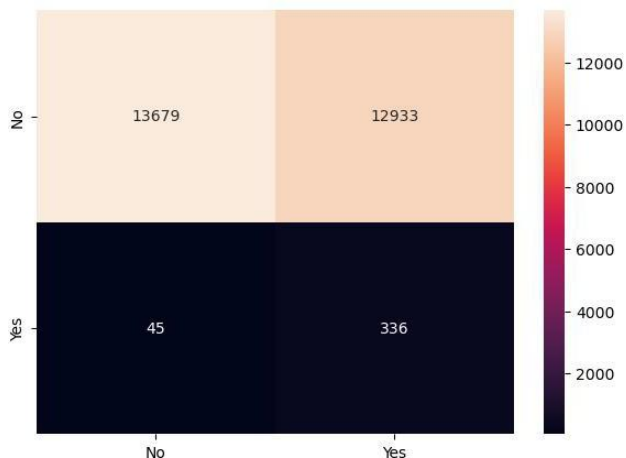


Figure 19: Bagging Confusion Matrix

We have used Bagging ensemble technique with Gaussian NB as the base model and number of estimators as 100

- Accuracy: 51.92%
- Recall: 88%
- Precision: 2.5%

Adaptive Boosting (Ada -Boosting)

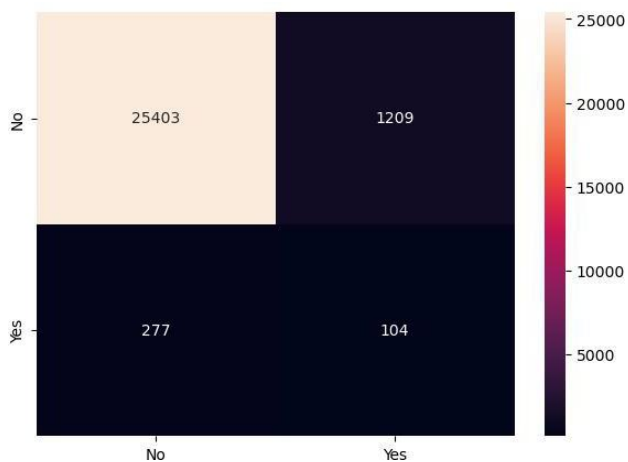


Figure 20: Ada BoostingConfusion Matrix

We have used Adaptive Boosting ensemble technique with Gaussian NB as the base model and number of estimators as 100

- Accuracy: 94.49%
- Recall: 27%
- Precision: 8%

Gradient Boosting

We have used Gradient Boosting ensemble technique with Gaussian NB as the base model and number of estimators as 100

- Accuracy: 94.78%
- Recall: 27%
- Precision: 8%

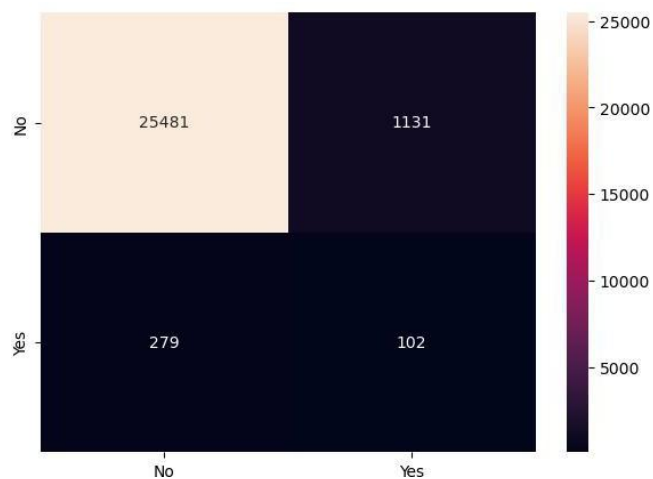


Figure 21: Gradient Boosting Confusion Matrix

5. Model Building and Tuning

Bootstrap Aggregation Hypermeter tuning

We have used Hypermeter tuning to determine optimal number of estimators that improves recall score

- n_estimators: 100
- Recall: 88.18%

Adaptive Boosting Hypermeter tuning

We have used Hypermeter tuning to determine optimal number of estimators that improves recall score

- n_estimators: 10
- Recall: 9.71%

Gradient Boosting Hypermeter tuning

We have used Hypermeter tuning to determine optimal number of estimators that improves recall score

- n_estimators: 10
- Recall: 61.42%

AUC Scores

Area Under Curve (AUC) scores for various models are as follows:

- Logistic Regression: 0.776
- LDA: 0.762
- Gaussian NB: 0.782
- KNN: 0.628
- Decision Tree: 0.562
- Bagging: 0.782
- Ada Boosting: 0.67
- Gradient Boosting: 0.784

6. Model Validation

Performance Summary

Below measures are used for evaluating performance of all the models on testing set..

- Accuracy: This denotes percentage of total sample being correctly identified (Defaulter/Non defaulter)
- Recall: This is the most important metric given the context of the study.This denotes percentage of actual defaulters being correctly identified as defaulters
- Precision: This denotes percentage of total predicted defaulters who are actually defaulters

Model	Accuracy	Recall	Precision
Logistic Regression	98.59%	0%	Undefined
LDA	97.89%	6%	10%
Gaussian NB	97.07%	13%	9%
KNN	98.43%	9%	31%
Decision Tree	97.55%	14%	14%

Table 2: Model Performance Summary

Interpretation

- As we can observe, recall value has suffered for all the models. Highest recall value is for Decision tree model (14%)
- However, given the context of our study, recall is a non-negotiable metric as it denotes the models capability to correctly predict a defaulter as a defaulter which is the sole purpose of our study.
- Hence, we can not finalize our model with a 14% of recall value.
- The reason of this poor performance in recallmetric is, the given dataset is highly imbalanced as 99% of the records fall into non-defaulter category.
- To develop a reliable default prediction model we need to have a balanced representation from both defaulter and non-defaulter class
- We use SMOTE to synthetically oversample the defaulter class to make the dataset balanced.

6. Model Validation

SMOTE

We have used SMOTE technique to synthetically oversample the defaulter class to make the training set balanced across defaulter and non-defaulter class.

- Before SMOTE Training set constitutes 62076 non-defaulters and 907 defaulters
- After SMOTE Training set constitutes 62076 samples for both defaulter and non-defaulters
- Performance metrics for the predictor models are depicted below after applying SMOTE

Model	Accuracy	Recall	Precision
Logistic Regression	64.77%	78%	3%
LDA	97.61%	9%	10%
Gaussian NB	51.92%	88%	3%
KNN	88.17%	30%	4%
Decision Tree	96.97%	13%	9%

Table 3: Model Performance Summary after SMOTE

Interpretation

- We can observe a significant jump in recall value for Gaussian NB (88%) and Logistic Regression (78%) models.
- However, we chose Gaussian NB for its highest ability of predicting the defaulter class accurately.
- Although accuracy and precision are significantly low for Gaussian NB but we decide to overlook this limitation given the context of the study.
- Gaussian NB model is used as the base estimator for our subsequent ensembling techniques.

7. Final Recommendation

Interpretation

- Based on our analysis, we have picked Gaussian NB model as a reliable default predictor that correctly identifies a defaulter 88% of times.
- We could observe some interesting trend for various features across defaulter and non-defaulter categories.
- Average amount spent on credit cards within 24 months in past from present to 12 months in past from present for defaulter class is 2035 whereas for non-defaulter class, it is 1516. Hence, more spend can lead to eventual default.
- Average age for non-defaulters is about 36 whereas for defaulters, average age is about 31.
- Clearly, average payment span in last 1 year for defaulters is 16 days whereas for non-defaulters is about 14 days. Hence, in last 1 year, defaulter has taken higher time to settle bills than non-defaulters.
- Maximum paid invoice in last 1 years or 2 years by non-defaulters is 40-50% higher than defaulters.
- Interestingly, Leisure and F&B are the 2 merchant groups where defaulters are more common than non-defaulters.